

Dr. Farhanul Hasan

POSTDOCTORAL FELLOW · SPACE TELESCOPE SCIENCE INSTITUTE

Muller 308 · 3700 San Martin Dr · Baltimore, MD-21218

✉ fhasan@stsci.edu | 🌐 <https://farhantasy.github.io/>

Education

New Mexico State University

Las Cruces, NM

PH.D.: ASTRONOMY; DEGREE RECEIVED: AUG 2024; ADVISOR: JOSEPH BURCHETT

Aug. 2018 - Aug. 2024

THESIS: "ON THE EVOLUTION OF GALAXIES IN COSMIC ECOSYSTEMS"

M.SC.: ASTRONOMY; DEGREE RECEIVED: AUG. 2022; ADVISORS: JOSEPH BURCHETT & CHRIS CHURCHILL

Aug. 2018 - Aug. 2022

Reed College

Portland, OR

B.A.: PHYSICS; DEGREE RECEIVED: MAY 2018; THESIS ADVISORS: ALISON CROCKER & JOHNNY POWELL

Aug. 2014 - May 2018

THESIS: "INTO THE VOID: MASS FUNCTION OF SUPERMASSIVE BLACK HOLES IN THE LOCAL UNIVERSE"

Experience

Space Telescope Science Institute

Baltimore, MD

POSTDOCTORAL RESEARCH FELLOW

Sept 2024 - Present

- Processing and analyzing grism spectroscopy and imaging data from James Webb and Hubble Space Telescopes
- Studying the physical properties of intermediate- and high-redshift galaxies, including metallicities and star formation rates
- Mapping the cosmic web, studying its gas properties and interplay with galaxy evolution, and making predictions for future galaxy surveys

New Mexico State University

Las Cruces, NM

GRADUATE RESEARCH ASSISTANT

Aug 2018 - Aug 2024

- Analyzing observed quasar spectra and cosmological simulations
- Developing new tools for large-scale structure identification from simulated and observational data
- Observing with ground-based telescopes at Apache Point and Keck Observatories

SCIENCE DEVELOPMENT LEAD, *CosmoVis*

May 2021 - Aug 2024

- Developing and improving *CosmoVis*, an open-source astrophysical 3D data visualization tool for exploring and analyzing hydrodynamical cosmological simulations of the universe, including identifying and implementing scientific uses and elucidating observations

GRADUATE TEACHING ASSISTANT

Aug. 2018 - Dec. 2022

Including ASTR 1115G/110G: Introduction to Astronomy, ASTR 305V: Life in the Universe, and ASTR 105G: The planets

Reed College

Portland, OR

UNDERGRADUATE RESEARCH ASSISTANT

May 2017 - Aug 2018

Analyzing large SDSS photometric and spectroscopic catalogs and photometric images of nearby spiral galaxy NGC 4736

University of California, Irvine

Irvine, CA

UNDERGRADUATE RESEARCH ASSISTANT

June 2016 - Aug. 2017

Analyzing space-based spectroscopic galaxy data, obtaining spectral energy distributions from UV to IR and measuring physical properties

Selected Publications

Hasan, F., et al., "Impact of Cosmic Filaments on Galaxy Morphological Evolution and Predictions of Early Cosmic Web Structure for Roman" *The Astrophysical Journal*, Volume 1004, Number 1, Page 74, Jun. 2026.

Luber, N., **Hasan, F., et al. "CHILES IX: Observational and Simulated HI Content and Star Formation of Blue Galaxies in Different Cosmic Web Environments" *The Astrophysical Journal*, Volume 985, Number 2, Page 214, May 2025.**

Hasan, F., et al., "Filaments of The Slime Mold Cosmic Web And How They Affect Galaxy Evolution" *The Astrophysical Journal*, Volume 970, Number 2, Page 177, Aug 2024.

Hasan, F., et al., "The Evolving Effect Of Cosmic Web Environment On Galaxy Quenching" *The Astrophysical Journal*, Volume 950, Number 2, Page 114, Jun 2023.

Hasan, F., et al., "Evolution of CIV Absorbers. II. Where Does CIV Live?" *The Astrophysical Journal*, Volume 924, Number 1, Page 12, Jan 2022.

Hasan, F., et al., "Evolution of CIV Absorbers. I. The Cosmic Incidence" *The Astrophysical Journal*, Volume 904, Number 1, Page 44, Nov 2020.

Acharyya, A.,..., **Hasan, F.**, et al., “**Spatially resolved gas-phase metallicity at $z \sim 2 - 3$ with JWST/NIRISS**”. accepted *Astronomy & Astrophysics*. Jun. 2026. arXiv:2508.05335v3.

Goovaerts, I.,..., **Hasan, F.**, et al., “**MXDFz4.4: A LyC emitter 250 Myr after the epoch of reionization and a first test of Ly α morphology as a tracer of LyC escape at high redshift**”. accepted, *Astrophysical Journal*. Jun. 2026.

Malkan, M.,..., **Hasan, F.**, et al., “**Parallel Application of Slitless Spectroscopy to Analyze Galaxy Evolution (PASSAGE): Survey Overview**”. *The Astrophysical Journal*, Volume 993, Number 1, Page 152, Oct 2025.

Runnholm, A.,..., **Hasan, F.**, “**The JWST/PASSAGE Survey: Testing Reionization Histories with JWST’s First Unbiased Survey for Lyman alpha Emitters at Redshifts 7.5-9.5**” *The Astrophysical Journal*, Volume 984, Number 1, Page 95, May 2025.

Huberty, M.,..., **Hasan, F.**, et al., “**The NIRISS PASSAGE Spectroscopic Redshift Catalog in COSMOS**”. in review, *Astrophysical Journal Supplements*. Jan. 2026. arXiv:2601.02220.

Nandi, G., Guo, Y., **Hasan, F.**, et al., “**Frozen in Size: Rest-frame Near-infrared Size Evolution of Quiescent and Star-forming Galaxies since Redshift $z = 4$ with JWST**”. in review, *Astrophysical Journal*. Apr. 2026.

Woo, J., Cowan, M., **Hasan, F.**, et al., “**The Cosmic Web Origins of Blue-Centred Galaxies**”. in review, *Nature Astronomy*. Jul. 2025.

Awards & Grants

2025-	PI: NASA High-End Computing Allocation SMD-24-54390923 , NASA Advanced Supercomputing Center	<i>Mountain View, CA</i>
2024-	Science PI: Director’s Discretionary Research Award , Space Telescope Science Institute	<i>Baltimore, MD</i>
2024-	Science PI: HST Cycle 31 Archival Research Award #17568 , Space Telescope Science Institute	<i>Baltimore, MD</i>
2023	Arts & Sciences Travel Grant , College of Arts & Sciences, New Mexico State University	<i>Las Cruces, NM</i>
2021-22	Merit-based Enhancement Award , Graduate School, New Mexico State University	<i>Las Cruces, NM</i>
2021	Zia Award for Outstanding Research by a Graduate Student , Dept. of Astronomy, NMSU	<i>Las Cruces, NM</i>
2019-20	College of Arts & Sciences General Scholarship , New Mexico State University	<i>Las Cruces, NM</i>
2018-20	William Webber Voyager Fellowship , Dept. of Astronomy, NMSU	<i>Las Cruces, NM</i>
2018	Commendation for Academic Excellence , Reed College	<i>Portland, OR</i>
2017	Research Fellowship Award , Reed College	<i>Portland, OR</i>
2017	Research Opportunity Grant , Reed College	<i>Portland, OR</i>
2016	Summer Experience Award , Reed College	<i>Portland, OR</i>

Selected Invited Talks

PASSAGE ISSI Collaboration Meeting	<i>Bern, Switzerland</i>
“GALAXY MORPHOLOGIES AND THEIR RELATIONSHIP WITH MASS, STAR FORMATION, AND METALLICITY AT COSMIC NOON”	<i>Feb. 2026</i>
University of Missouri, Columbia Astrophysics Seminar	<i>Columbia, MO</i>
“GALAXY EVOLUTION IN THE COSMIC WEB: FROM LARGE TO SMALL SCALES OF COSMIC ECOSYSTEMS”	<i>Sep. 2025</i>
2025 Santa Cruz Galaxy Workshop	<i>Santa Cruz, CA</i>
“THE EARLY COSMIC WEB: IMPACT ON GALACTIC MORPHOLOGICAL EVOLUTION AND PREDICTIONS FOR ROMAN”	<i>Aug. 2025</i>
Baryon Pasting Collaboration 2025 meeting	<i>New Haven, CT</i>
“MODELING COSMIC WEB FILAMENTS: CHALLENGES, ADVANCEMENTS, AND FUTURE DIRECTIONS”	<i>Apr. 2025</i>
Yale University Special Galaxy Seminar	<i>New Haven, CT</i>
“COSMIC FILAMENTS: FRIENDS OF FOE OF GALAXY EVOLUTION?”	<i>Apr. 2025</i>
College University of New York (NYC College of Technology) Theoretical Physics Seminar	<i>New York, NY</i>
“THE SLIME MOLD COSMIC WEB AND WHAT IT TEACHES US ABOUT GALAXY EVOLUTION”	<i>Nov. 2024</i>
Princeton University Galaxy Discussion Journal Club (Galread)	<i>Princeton, NJ</i>
“FILAMENTS OF THE SLIME MOLD COSMIC WEB AND HOW THEY AFFECT GALAXY EVOLUTION”	<i>Oct. 2024</i>
2024 Santa Cruz Galaxy Workshop	<i>Santa Cruz, CA</i>
“FILAMENTS OF THE SLIME MOLD COSMIC WEB”	<i>Aug. 2024</i>
Steward Observatory/NOIRLab Galaxy Group	<i>Tucson, AZ</i>
“A COSMIC CONNECTION: HOW GALAXY FORMATION IS AFFECTED BY THE COSMIC WEB”	<i>Oct. 2023</i>
Space Telescope Science Institute/Johns Hopkins University Galaxies & AGN Journal Club	<i>Baltimore, MD</i>
“A COSMIC CONNECTION: HOW GALAXY FORMATION IS AFFECTED BY THE COSMIC WEB”	<i>Sep. 2023</i>

Columbia University Seminar

"A COSMIC CONNECTION: HOW GALAXY FORMATION IS AFFECTED BY THE COSMIC WEB"

New York, NY

Sep. 2023

Yale University Galaxy Lunch

"A COSMIC CONNECTION: HOW GALAXY FORMATION IS AFFECTED BY THE COSMIC WEB"

New Haven, CT

Sep. 2023

Arizona State University Borthakur Extragalactic Group

"THE IMPACT OF COSMIC WEB STRUCTURES ON GALAXY FORMATION"

Tempe, AZ (Virtual)

Sep. 2022

2022 Santa Cruz Galaxy Workshop

"THE IMPACT OF COSMIC WEB STRUCTURES ON GALAXY FORMATION"

Santa Cruz, CA

Aug. 2022

Selected Contributed Talks

The Baryon Cycle from Reionization to the Cosmic Noon

"THE EVOLVING ROLE OF THE COSMIC WEB IN GROWING AND SHAPING GALAXIES"

Puerto Varas, Chile

Dec. 2025

"Some Like It Hot!" A Journey from the Hot IGrM to the Multiphase CGM

"A COSMIC CONNECTION: THE IMPACT OF THE COSMIC WEB ON GALAXY EVOLUTION"

Garching, Germany

May 2025

243rd Meeting of the American Astronomical Society

"A COSMIC CONNECTION: THE EFFECT OF THE COSMIC WEB ON GALAXY EVOLUTION" (DISSERTATION TALK)

New Orleans, LA

Jan. 2024

Oases in the Cosmic Desert: Understanding the Structure of the Circumgalactic Medium

"THE CONNECTION BETWEEN GALAXY QUENCHING AND COSMIC WEB STRUCTURES ACROSS COSMIC TIME"

Tempe, AZ

Feb. 2023

The Co-evolution of the Cosmic Web and Galaxies across Cosmic Time at KITP

"HOW DO COSMIC WEB STRUCTURES AFFECT GALAXY QUENCHING ACROSS COSMIC TIME?"

Santa Barbara, CA

Feb. 2023

Service and Leadership

Guest Lecturer, JHU pre-college "Exploring the universe with Space Telescopes" program

JOHNS HOPKINS UNIVERSITY

Baltimore, MD

Jun 2026 – Aug 2026

Local Organizing Committee, STScI Spring Symposium 2026

SPACE TELESCOPE SCIENCE INSTITUTE

Baltimore, MD

Jan 2026 – Present

Panel Support Scientist, JWST Cycle 5 Peer Review

SPACE TELESCOPE SCIENCE INSTITUTE

Baltimore, MD

Nov 2025 – Feb 2026

HotSci 2025 Summer Colloquium Series Organizer

SPACE TELESCOPE SCIENCE INSTITUTE

Baltimore, MD

Apr 2025 – Aug 2025

Postdoc Representative to the Science Mission Office

SPACE TELESCOPE SCIENCE INSTITUTE

Baltimore, MD

Feb 2025 – Present

Journal Paper Referee

APJ, A&A, MNRAS, OJAA, RASTI

Virtual

May 2023 – Present

Co-supervisor/Research Mentor

DEPARTMENT OF ASTRONOMY, NEW MEXICO STATE UNIVERSITY

Las Cruces, NM

May 2022 - Aug. 2024

Executive Secretary for NASA Astrophysics Proposal Review Panel

NASA ROSES

NASA

May 2023

President, Bangladesh Student Association (BSA)

NEW MEXICO STATE UNIVERSITY

Las Cruces, NM

Aug. 2023 - Aug. 2024

Mentor for Prospective Graduate Students

CIENTÍFICO LATINO GRADUATE STUDENT MENTORSHIP INITIATIVE (GSMI)

Científico Latino

Jul. 2020 - May 2022

Community Outreach Volunteer

DEPARTMENT OF ASTRONOMY, NEW MEXICO STATE UNIVERSITY

Las Cruces, NM

Aug. 2018 - Aug. 2024

Co Mentor, UC CROSS

GOOGLE SUMMER OF CODE 2022

UC Santa Cruz

Jun. 2022-Dec. 2022

Secretary, Astronomy Graduate Student Organization (AGSO)

DEPARTMENT OF ASTRONOMY, NEW MEXICO STATE UNIVERSITY

Las Cruces, NM

Aug. 2019 - May 2020